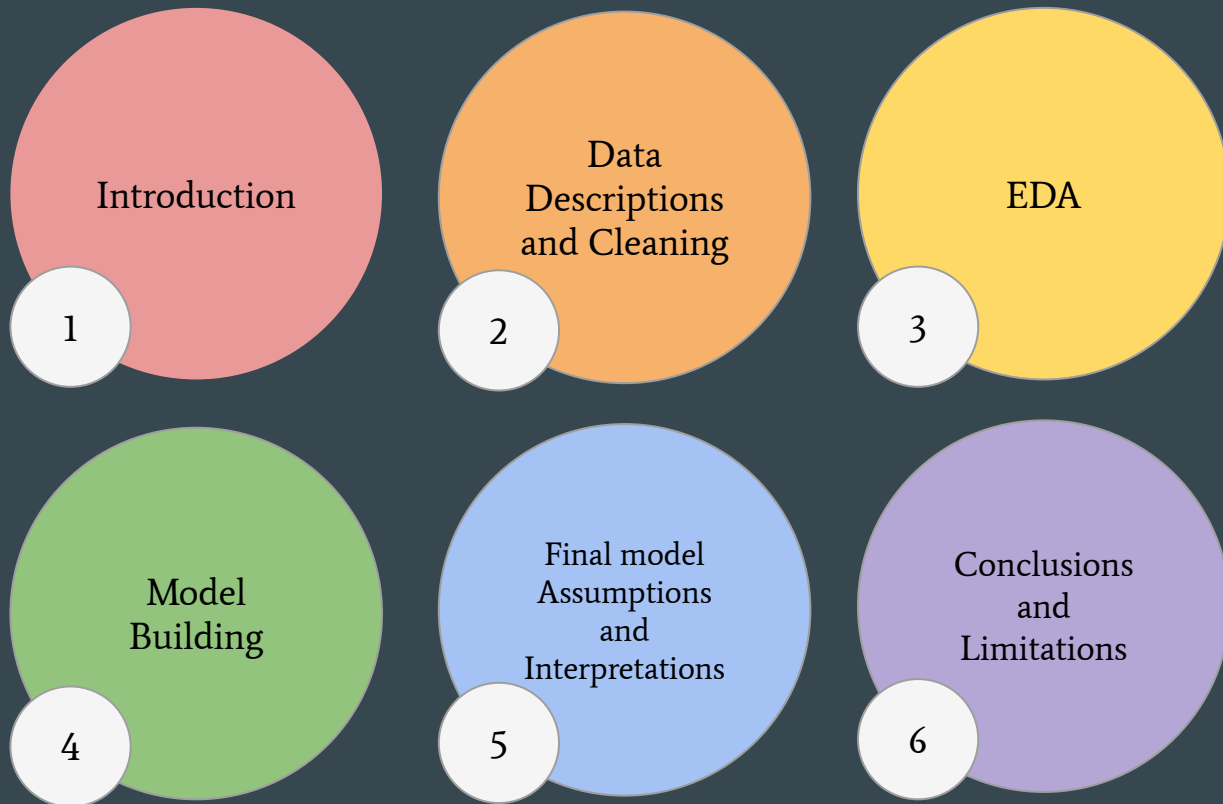


My Stat 183 Project: Housing Prices in California

...

By: Cameron Fong

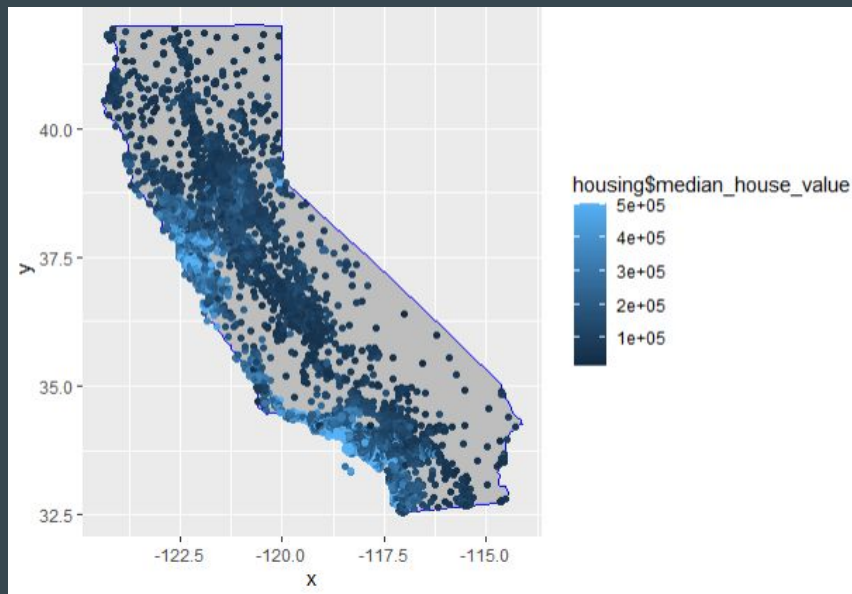
Overview



Introduction

Question of Interest: What factors influence the prices of a house in California and how much of an effect do they have?

Reasoning: With house prices skyrocketing in prices especially in California, I would like to explore what causes specific houses to be more expensive or cheaper than others so that I may take them into consideration for the future when the time comes to buy one for myself



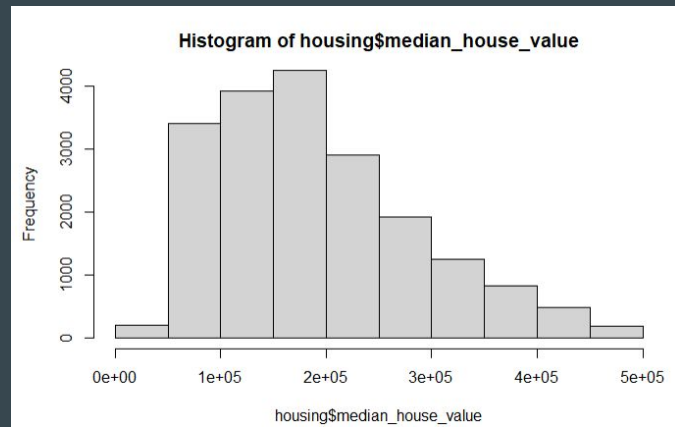
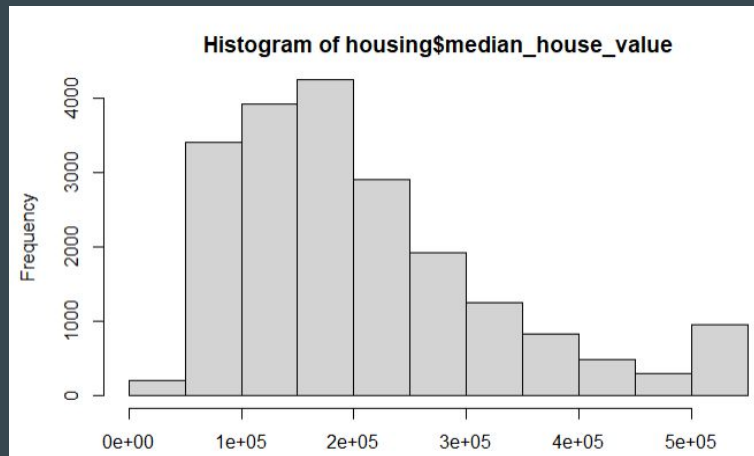
Data Cleaning

Data set: Housing

- 20640 observations (houses), 10 variables
- Dataset was obtained from kaggle
- All houses are from California's 1990 census

Cleaning the data

- Removing all houses with any NA values
 - 20640 to 20433 observations
- Removing all outliers within response variable `median_house_value` (Y) using IQR
 - All `median_house_values` greater than 482,500 removed: 20433 to 19369 observations
 - Second removal of outliers in response variable: 19369 to 19337 observations (shown later)



Data Descriptions

Response variable (1)

- Median_house_value (Numerical)

Predictor variables (9)

- Longitude (Numerical)
- Latitude (Numerical)
- Housing_median_age (Numerical)
- Total_rooms (Numerical)
- Total_bedrooms (Numerical)
- Population (Numerical)
- Households (Numerical)
- Median_income (Numerical)
- Ocean Proximity (Categorical)

A tibble: 6 x 10

longitude <dbl>	latitude <dbl>	housing_median_age <dbl>	total_rooms <dbl>	total_bedrooms <dbl>	population <dbl>	households <dbl>	median_income <dbl>	median_house_value <dbl>	ocean_proximity <chr>
-122.23	37.88	41	880	129	322	126	8.3252	13.02276	NEAR BAY
-122.22	37.86	21	7099	1106	2401	1138	8.3014	12.78968	NEAR BAY
-122.24	37.85	52	1467	190	496	177	7.2574	12.77167	NEAR BAY
-122.25	37.85	52	1274	235	558	219	5.6431	12.74052	NEAR BAY
-122.25	37.85	52	1627	280	565	259	3.8462	12.74315	NEAR BAY
-122.25	37.85	52	919	213	413	193	4.0368	12.50507	NEAR BAY

EDA - Response Variable

My response variable is median house pricing

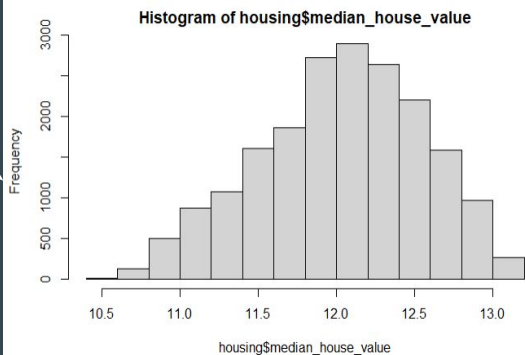
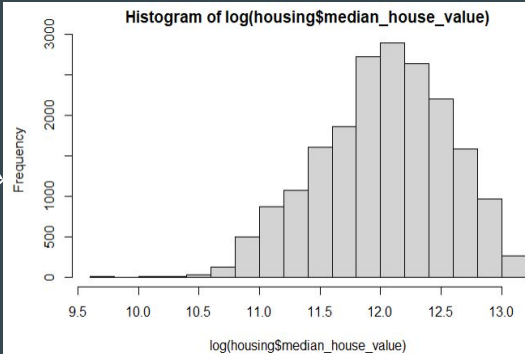
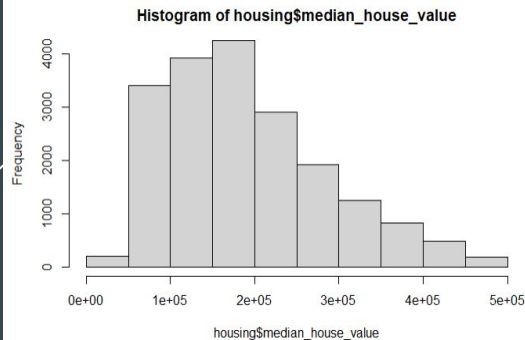
- Median house value within a block in dollars
- Ranges from 14,999 to 482,200

I to II: Median house pricing has a slight right skew

- Solution: log transformation

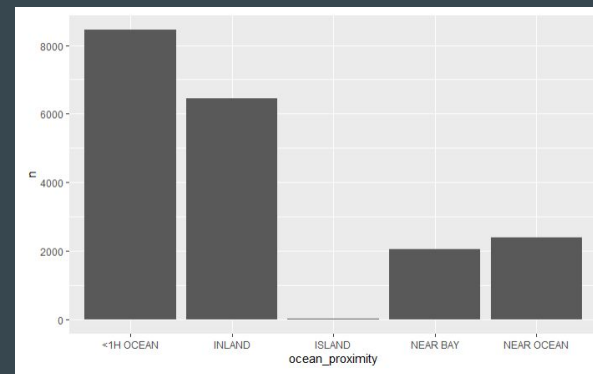
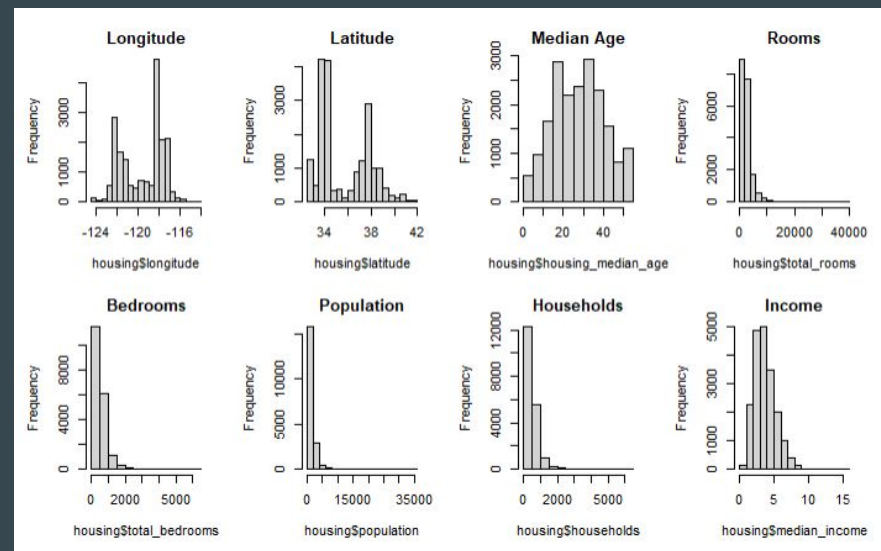
II to III: Median house pricing after log transformation now has a left skew

- Solution: eliminate outliers using IQR



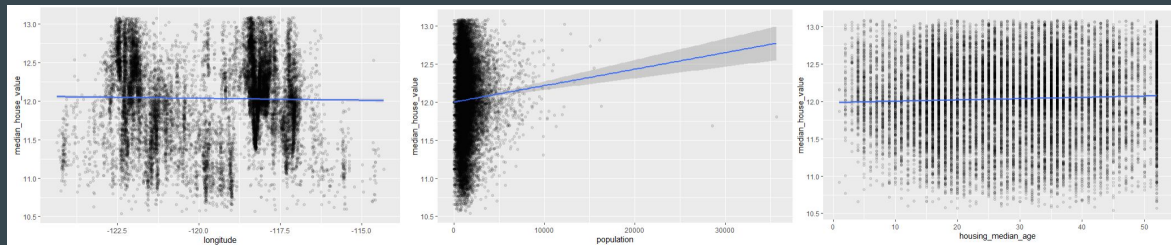
EDA - Predictor Variables

- Longitude: specifies east and west position of a block in California
 - Ranges from -124.3 to -114.3
- Latitude: Specifies north to south position of a block in California
 - Ranges from 32.54 to 41.95
- Housing_median_age: Median age of a house in a block in years
 - Ranges from 1 to 52 years
- Total_rooms: Total amount of rooms in a block
 - Ranges from 2 to 39320
- Total_bedrooms: Total amount of bedrooms in a block
 - Ranges from 2 to 6645
- Population: Total number of people living within a block
 - Ranges from 3 to 35682
- Households: Total number of households in a block
 - Ranges from 2 to 6082
- Median_income
 - Measured in tens of thousands of dollars
 - Ranges from 0.5 to 15 thus 5000 to 150,000 dollars
- Ocean Proximity
 - How close a block of houses is to the ocean
 - Categorized by island, near bay, near ocean, inland, and <1h from ocean



<1H OCEAN	INLAND	ISLAND	NEAR BAY	NEAR OCEAN
8451	6439	5	2053	2389

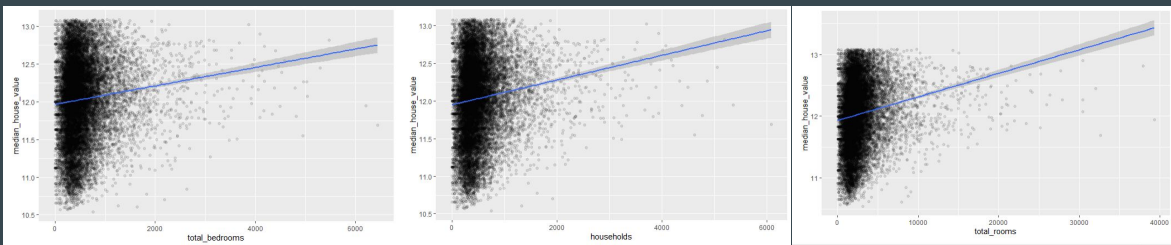
EDA - Response vs predictors



Longitude Correlation: -0.018

Population Correlation: 0.047

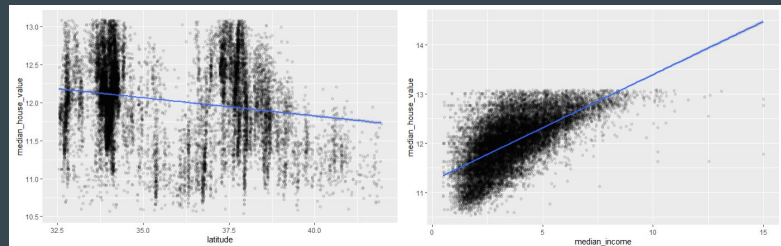
Median Age Correlation: 0.041



Bedrooms Correlation: 0.098

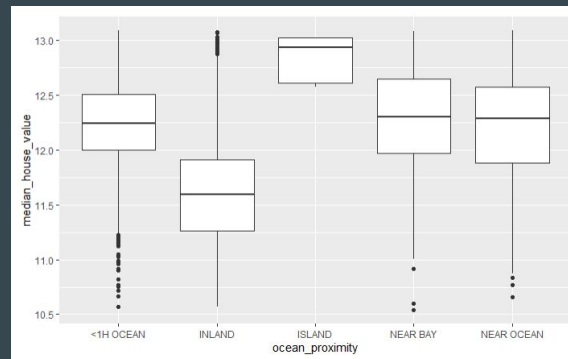
Households Correlation: 0.119

Rooms Correlation: 0.159



Latitude Correlation: -0.196

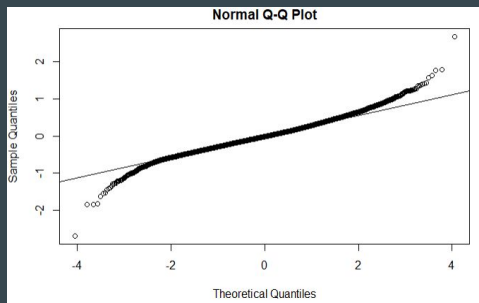
Income Correlation: 0.641



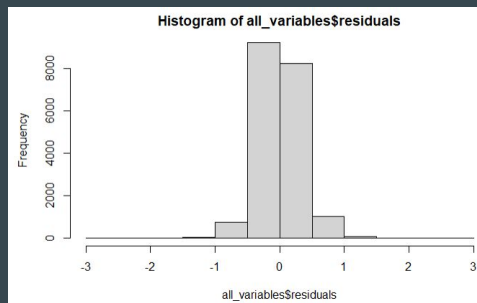
Boxplot: Inland is notably cheaper housing compared to all other categorical variables

Model building

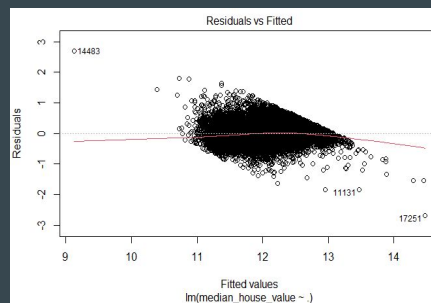
```
all_variables = lm(median_house_value ~., data = housing)
```



QQplot looks approximately linear



Residuals are normal



Residuals vs fitted has a majority of the data distributed on the 0 line thus a linear model is appropriate

```
Durbin-watson test
data: all_variables
Dw = 0.93855, p-value < 2.2e-16
```

Durbin watson test fails as there is positive autocorrelation and thus the potential for our significant predictor values to not be significant

Stepwise regression

- All predictors are kept (9 variables total)
- Gained an AIC of -141478.3
- All variables have a p-value less than 0.05 thus are significant to median_house_value

```
Step: AIC=-141478.3
log(median_house_value) ~ median_income + ocean_proximity + total_bedrooms +
  population + housing_median_age + total_rooms + households +
  longitude + latitude
```

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)
<none>			12.832	-141478		
~ households	1	0.0150	12.848	-141478	22.614	1.994e-06 ***
~ total_rooms	1	0.0687	12.901	-141377	103.490	< 2.2e-16 ***
~ housing_median_age	1	0.1169	12.949	-141305	176.032	< 2.2e-16 ***
~ total_bedrooms	1	0.1618	12.994	-141238	243.698	< 2.2e-16 ***
~ population	1	0.4961	13.329	-140747	747.095	< 2.2e-16 ***
~ latitude	1	0.6276	13.460	-140557	945.142	< 2.2e-16 ***
~ longitude	1	0.6645	13.497	-140504	1000.588	< 2.2e-16 ***
~ ocean_proximity	4	0.9331	13.766	-140129	351.299	< 2.2e-16 ***
~ median_income	1	6.6663	19.499	-133390	10038.651	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model building - Vif Test

The stepwise regression and residuals both show that the full model is the optimal choice to predicting our response variable median house price, but there are other conditions that must be considered, namely vif to make sure the predictor estimates are accurate.

```
lm_test = lm(value~income + longitude + latitude + medianage + rooms + bedrooms + population + households + ocean)
vif(lm_test)
```

	GVIF	Df	GVIF^(1/(2*Df))
income	1.874617	1	1.369167
longitude	17.823658	1	4.221807
latitude	19.801924	1	4.449935
medianage	1.369681	1	1.170334
rooms	14.019392	1	3.744248
bedrooms	36.878074	1	6.072732
population	6.563384	1	2.561910
households	34.444370	1	5.868933
ocean	4.048221	4	1.190990

Longitude, latitude, rooms, bedrooms, and households all have vif values greater than 10, meaning that the variance of these specific parameter estimates are inflated, thus are very unstable and should not be trusted.

Model building - VIF part 2

Linear Model	VIF	Variable(s) removed																																				
Value~income + longitude + latitude + medianage + rooms + population + households + ocean	<table><thead><tr><th></th><th>GVIF</th><th>Df</th><th>GVIF^(1/(2*Df))</th></tr></thead><tbody><tr><td>income</td><td>1.595585</td><td>1</td><td>1.263165</td></tr><tr><td>longitude</td><td>17.655829</td><td>1</td><td>4.201884</td></tr><tr><td>latitude</td><td>19.706737</td><td>1</td><td>4.439227</td></tr><tr><td>medianage</td><td>1.359938</td><td>1</td><td>1.166164</td></tr><tr><td>rooms</td><td>10.082297</td><td>1</td><td>3.175263</td></tr><tr><td>population</td><td>6.133537</td><td>1</td><td>2.476598</td></tr><tr><td>households</td><td>12.340028</td><td>1</td><td>3.512838</td></tr><tr><td>ocean</td><td>4.004309</td><td>4</td><td>1.189367</td></tr></tbody></table>		GVIF	Df	GVIF^(1/(2*Df))	income	1.595585	1	1.263165	longitude	17.655829	1	4.201884	latitude	19.706737	1	4.439227	medianage	1.359938	1	1.166164	rooms	10.082297	1	3.175263	population	6.133537	1	2.476598	households	12.340028	1	3.512838	ocean	4.004309	4	1.189367	Bedrooms
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VIF condition is met with the last linear model thus we stop there

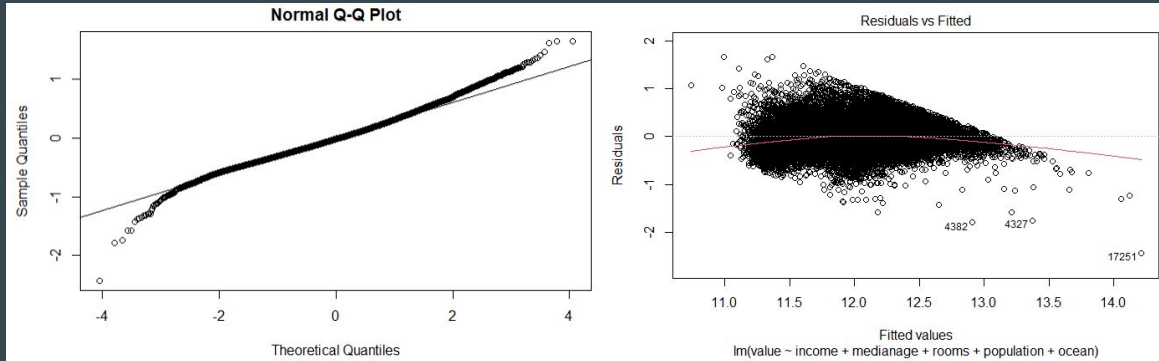
Model Assumptions

To ensure the validity and reliability of a linear model, we check that the model has:

- Normality
- Constant variance
- Independence
- No outliers

We assume all blocks in California are independent and we removed any blocks with outliers in our response variable `median_house_value`, thus we have normality and constant variance to check

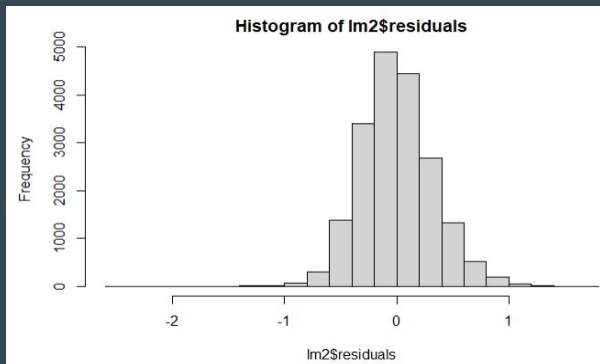
Normality



Normality is met

- Data fits a linear line (QQplot)
- Fitted values mostly revolve around the 0 residual line

Variance



Constant variance is met

- Histogram of residuals is approximately normal and centered around 0

Durbin-Watson test

```
data: lm2
DW = 0.79157, p-value < 2.2e-16
```

Low durbin-watson test statistic

- Positive autocorrelation
- Potentially predictors are significant when in reality they are not

Final Model Interpretations

Final model: $\log(\text{median_house_price}) = 11.43 + 0.1741 \cdot \text{income} + 0.003085 \cdot \text{medianage} + 0.00005171 \cdot \text{rooms} - 0.00006959 \cdot \text{population} - 0.4956 \cdot \text{Inland} + 0.7556 \cdot \text{Island} + 0.002516 \cdot \text{Bay} + 0.0273 \cdot \text{Ocean}$

- For every increase in 10,000 for average median income, log median house prices increase by 0.1741 dollars on average
- For every year increase in median age of the people in a block, log median house prices increases by 0.003085 dollars
- For every 1 room increase in a block, log median house prices increases by 0.00005171 dollars
- For every 1 person increase in a bloc, log median house prices increases by 0.00006959 dollars
- If the block is considered inland, log median house prices decreases by 0.4956 dollars
- If the block is considered on an island, log median house prices increases by 0.7556 dollars
- If the block is considered near the bay, log median house prices increases by 0.002516 dollars
- If the block is considered near the ocean, log median house prices increases by 0.0273 dollars

Residuals:

Min	1Q	Median	3Q	Max
-2.42516	-0.21587	-0.01716	0.19813	1.65291

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.143e+01	1.182e-02	967.191	< 2e-16 ***
income	1.741e-01	1.719e-03	101.296	< 2e-16 ***
medianage	3.085e-03	2.157e-04	14.304	< 2e-16 ***
rooms	5.171e-05	2.347e-06	22.036	< 2e-16 ***
population	-6.959e-05	4.324e-06	-16.091	< 2e-16 ***
oceanINLAND	-4.956e-01	5.820e-03	-85.154	< 2e-16 ***
oceanISLAND	7.556e-01	1.450e-01	5.211	1.9e-07 ***
oceanNEAR BAY	2.516e-03	8.219e-03	0.306	0.759561
oceanNEAR OCEAN	2.730e-02	7.572e-03	3.605	0.000313 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.324 on 19328 degrees of freedom
Multiple R-squared: 0.6177, Adjusted R-squared: 0.6175
F-statistic: 3903 on 8 and 19328 DF, p-value: < 2.2e-16

Conclusion

- Houses closer to the ocean are noticeably more expensive than those more in land
- Higher income, median age, population, and total rooms increases the median house price
 - Income had the largest impact on median house price
- Higher population decreases the median house price
- All variables are significant except for ocean near bay with a p-value of 0.759
- There is an adjusted R^2 of 0.6177 meaning that 61.77% of variability within median_house_price can be explained by our linear model
 - 0.6177 is fairly accurate for determining median house prices

Limitations

- 19,337 is a small portion of blocks compared to all blocks in California (over 700,000 in total) thus would be better to get a larger sample
- By removing all variables with NA and outliers, I could have lost out on important data that could better predict median_housing_prices
- Only getting data from blocks as a whole rather than by individual houses
- Residuals vs fitted still has a small negative slope rather than a constant variance due to a few outliers
- Not being able to get ocean proximity to fully pass the significance test

Future Work

- Testing interactions within the predictor variables to increase the adjusted R^2
- Getting a more recent census to get a more accurate prediction of house prices today
- Testing with more samples to get a better representation of the population of houses in California

Thank you